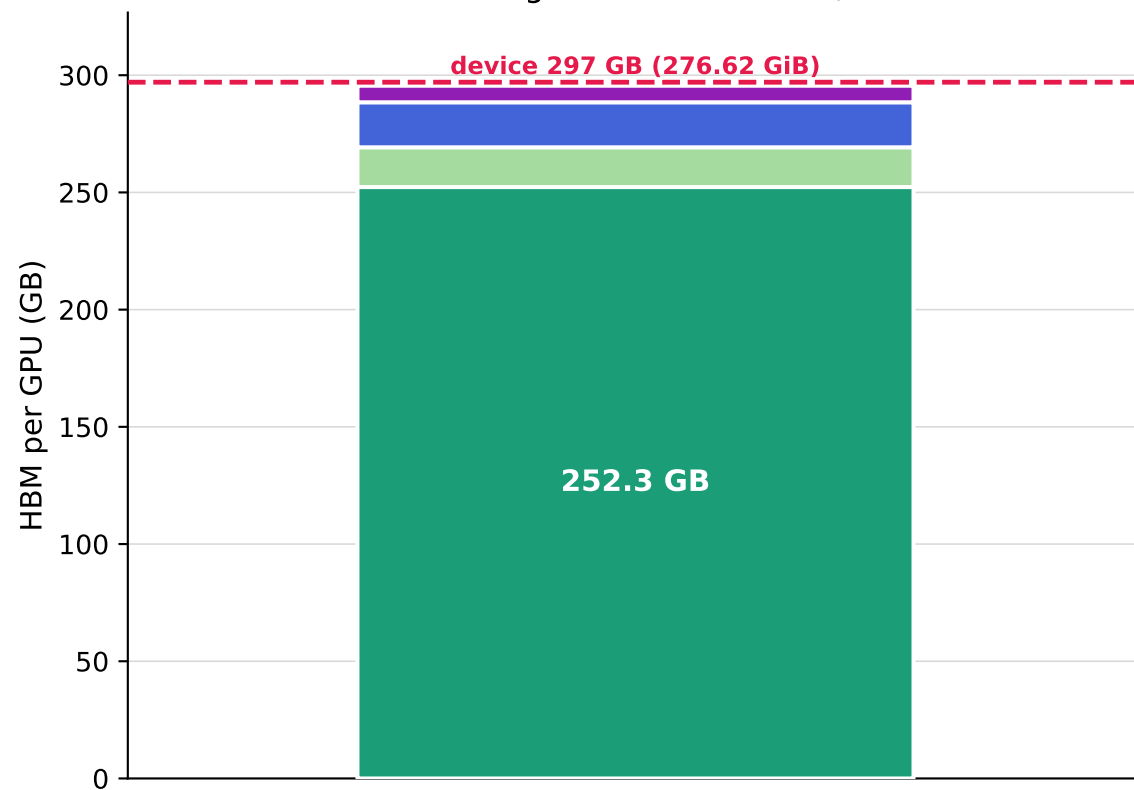


GB300 v26.06 — the HBM model, every term measured or exact (1B rows, 8 GPU, ratio 0.1)

Where the device goes — 1B/8/bs512, at the OOM



- PyTorch allocated (LIVE) — 252.3 GB
= $bs \times 0.522 + 3.5$ MEASURED
- fragmentation — 17.0 GB
MEASURED (16-17 GB at the limit)
- DynamicEmb cache — 19.2 GB
= $(R/N) \times ratio \times 1536$ B EXACT
- CUDA ctx + NCCL — 7.2 GB
MEASURED (7.2 @N=8 → 21.2 @N=32)
- free — 1.4 GB
needed 1.92 GiB → OOM

Only activations scale with batch — so the batch ceiling is arithmetic

